

SARTHAK SHRIVASTAVA

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EDUCATION

Indian Institute of Technology Delhi, India 2019 - 2023
Bachelor of Technology (B.Tech.), Civil Engineering GPA: 8.283/10.0

AREAS OF INTEREST

Computational Structural Mechanics, Machine Learning, Continuum Mechanics, Finite Element Methods, Multi-scale and Multi-physics Modeling, Peridynamics, Structural Dynamics, Structural Health Monitoring, Tribology

RESEARCH EXPERIENCE

- Undergraduate Researcher** | *Multiphysics & Multiscale Mechanics Research Group, IIT Delhi* (Sep, 2022 - Jun, 2023)
- Conducting numerical simulations using finite element and peridynamics methods to investigate the role of indenter geometry on indentation-induced damage of glasses under the supervision of Prof. N. M. Anoop Krishnan
- Summer Undergraduate Research Fellow** | *Industrial Research & Development Unit, IIT Delhi* (May, 2021 - Oct, 2021)
- Worked on a summer research project titled 'The Application of Machine Learning to Structural Health Monitoring' under the supervision of Prof. Sahil Bansal, funded by the Industrial Research & Development (IRD) Unit, IIT Delhi

PROJECTS

- Investigating Indentation Size Effect in Glasses using Computational Methods** (Sep, 2022 - Jun, 2023)
Advisor: Prof. N. M. Anoop Krishnan, IIT Delhi
- Conduct in-depth numerical simulations employing **finite element** and **peridynamics** techniques to analyze the influence of indenter geometry on glass **indentation-induced damage**
 - Develop a comprehensive damage model for glass indentation, including crack initiation, propagation, and fragmentation; investigate the **indentation size effect** by varying loads and indenter tip angles to understand glass behavior
 - Utilize **machine learning** to predict glass mechanical properties, particularly hardness, based on composition, indenter tip angle, and load, fostering a deeper understanding
- Stability of Steel Columns at Elevated Temperatures** | *Finite Element Analysis* (Nov, 2022 - Dec, 2022)
Advisor: Prof. Vasant Matsagar, IIT Delhi
- Formulated eigen-value problem for **linear bifurcation buckling** analysis; conducted **mesh convergence** studies
 - Explored **variational approach** for thermal force vector determination and studied stresses under **thermo-mechanical eccentric loading** for various boundary conditions
 - Performed **nonlinear analysis**, incorporating **Riks approach** for post-buckling until collapse due to thermal straining
- The Application of Machine Learning to Structural Health Monitoring** | *UROP* (May, 2021 - Oct, 2021)
Advisor: Prof. Sahil Bansal, IIT Delhi
- Devised a **vibration-based damage-detection** approach using **AI methods** on time response simulated data
 - Extracted different **damage-sensitive features** based on **signal statistics** and **modal properties**; analyzed **signal response time-series** using **AR and ARX models**; proposed **statistical model for feature discrimination**
 - Ranked damage-sensitive features and employed **support vector machines** to classify **structural condition states**
- Transverse Vibration Analysis of Euler-Bernoulli Beam** | *Computational Mechanics* (Mar, 2022 - Apr, 2022)
Advisor: Prof. Prapanch Nair, IIT Delhi
- Derived an **explicit finite difference scheme** with **second order accuracy** by discretizing the governing equations
 - Implemented Python script for computation of **natural frequencies** of vibration, **mode shapes**, and **vibration response** of the elastic beam for **different boundary conditions**, viz. fixed-fixed, fixed-pinned, pinned-pinned, and fixed-free
 - von Neumann analysis** was done to establish **stability requirements** of the solution, explicit FD scheme was found to be **conditionally stable** and **convergent**; compared the results with analytical solution to validate the approach
- Structural Design using STAAD.Pro** | *Structural Design & Detailing* (Aug, 2022 - Nov, 2022)
- Designed and analyzed the structures of **hostel campus** comprising five blocks of **G+10 storey buildings**, **reinforced concrete overhead water tank**, **industrial shed truss**, and **transmission tower** using AutoCAD and STAAD.Pro
 - Resistance to **earthquake** and **wind loads** were incorporated into the structural design process as per Indian Standards

PUBLICATIONS & PRESENTATIONS

- The Role of Indenter Geometry on Indentation-Induced Damage of Silica Glass using Peridynamics*, ACerS GOMD Annual Meeting, LA, United States. Jun 2023. [Abstract accepted]
- Impact Studies with Peridynamics*, Civil Engineering Seminar, Indian Institute of Technology, Delhi, India. Apr 2022.

TEACHING & MENTORING EXPERIENCE

Tutor | Course Hero, Inc.

(Dec, 2021 - Apr, 2023)

- Offered online tutoring to students seeking academic assistance in Physics, Mathematics, and Civil Engineering
- Received **consistently positive ratings** and feedback from students for effectiveness and clarity of instruction

Academic Mentor | Board of Student Welfare (BSW), IIT Delhi

(Nov, 2020 - Feb, 2021)

- Provided **peer mentorship** and tutoring to undergraduate freshers in **Engineering Mechanics** (APL100) course
- Assisted the students to acclimate to a new academic setting by **facilitating weekly study sessions**

Educator (IIT-JEE Physics) | Unacademy

(Aug, 2020 - Oct, 2020)

- Taught Advanced Physics to **500+ IIT-JEE aspirants**; held interactive sessions and emphasized **analytical thinking, problem-solving** and **exam-taking** skills through the use of **visual learning** approaches; offered learning resources
- Ranked among **top 100 educators** in IIT-JEE category with over **15K minutes** of watch time on online platforms

Mentor | Project Aarohan, National Service Scheme (NSS), IIT Delhi

(Sep, 2019 - Apr, 2020)

- Mentored underprivileged students of class XI and XII for competitive exams
- Taught Physics; provided them with comprehensive study material and conducted doubt clearing sessions

WORK EXPERIENCE

Machine Learning Engineer | Sirion

(Jul, 2023 - Present)

- Developing **AI-powered document understanding** solutions for Contract Lifecycle Management (CLM) using **Retrieval Augmented Generation (RAG)** for efficient context-aware processing
- Optimized information retrieval with **vector databases** and **fine-tuned embedding models** for domain-specific CLM corpus; evaluated and integrated **LLMs (GPT, Llama, Mistral, Gemma, etc.)** based on accuracy and performance
- Ensured scalability and efficient handling of large workloads by leveraging **multi-threaded message queuing**

Summer Analyst | Axis Bank (Business Intelligence Unit)

(Jun, 2022 - Jul, 2022)

- Built a **web-scraping tool** that extracts information related to employees' salaries, revenues, capitals of an organization
- Employed **Selenium** to **automate web browser** and interact with **dynamic web pages**; handled **captcha recognition** through **OpenCV** based **character segmentation** technique; analyzed the scraped data to gain useful **business insights**
- Developed tool is capable of scraping data of **600K+ organizations**; proposed ideas to **amplify sales** of salary accounts

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Java, Julia, MATLAB
- **Software & Tools:** Git, Docker, High Performance Computing, STAAD.Pro, Abaqus, Ansys, $\LaTeX$, Microsoft Office
- **Libraries:** PyTorch, OpenCV, Pandas, NumPy, SciPy, Matplotlib, Scikit-learn, Transformers, Peridigm, LangChain

HONORS & AWARDS

- Youngest recipient of **Sirion's DAO Award**, exemplifying Diversity, Agility, and Ownership with excellence 2023
- Awarded the **Summer Undergraduate Research Award (SURA)**, 2021 from Industrial R&D Unit, IIT Delhi 2022
- **Online Judges:** Scored **5-star rating** on **HackerRank** and **3-star (max. 1783 rating)** on **CodeChef** 2021
- **JEE (Advanced):** Secured **All India Rank 1841 (top 1 percentile)** among 0.17 million candidates 2019
- **JEE (Main):** Achieved **All India Rank 1403 (top 0.1 percentile)** among 1.2 million candidates 2019
- **KVPY:** Awarded fellowship for showcasing excellence in scientific aptitude and research potential by DST, GoI 2018

REFERENCES

Dr. N. M. Anoop Krishnan

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